DFT reference vs MLIP prediction | complete energy and force train/test parity

A Scratch-600K

B FT-600K

C Scratch-MultiT

D FT-MultiT

train

test

n=5,358
ρ=0.89, r=0.90

n=5,358
ρ=0.85, r=0.86

n=7,977
ρ=0.89, r=0.93

n=7,977
ρ=0.95, r=0.96

n=801
ρ=0.45, r=0.46

n=801
ρ=0.47, r=0.44

n=2,301
ρ=0.52, r=0.46

n=2,301
ρ=0.43, r=0.44

MLIP $E/N - \langle E/N \rangle_{\text{cluster}}$ (eV atom^{-1})

DFT $E/N - \langle E/N \rangle_{\text{cluster}}$ (eV atom^{-1})

Energy count per hexbin

n=1,336,032
ρ=0.99, r=0.99

n=1,336,032
ρ=1.00, r=1.00

n=1,983,330
ρ=0.99, r=0.99

n=1,983,330
ρ=1.00, r=1.00

n=206,658
ρ=0.84, r=0.80

n=206,658
ρ=0.88, r=0.88

n=584,658
ρ=0.86, r=0.74

n=584,658
ρ=0.90, r=0.90

MLIP force F (eV Å^{-1})

DFT force F (eV Å^{-1})

Force count per hexbin

train test